🟡 AlphaQuest Capital — BTC Signal Dashboard

FINTECH 717 — Investment Committee Pitch | Strategy: Risk-Managed Long/Cash with 5% Stop-Loss kill switch

🚀 Run Full Pipeline

✅ Pipeline executed successfully!

🔮 Tomorrow's Trading Signal

Today's BTC Close	Signal	Model Confidence	Prob. Cutoff
$93,530.23	BUY	51.0%	49%
↓ -1.72%		↑ above cutoff	

💰 $50M Capital Allocation — Performance Report

Simulated deployment of USD $50,000,000 into the Risk-Managed Long/Cash strategy on the 2024 test period.

📩 Capital Deployed	📩 Portfolio Value	📈 Strategy Return	⚡ Alpha vs Buy&Hold
$50M	$100.00M	+100.00%	-11.77%
↑ Starting capital	↑ +50.00M profit/loss	↑ vs Buy&Hold +111.76%	↑ excess return

📊 Sharpe Ratio	📉 Max Drawdown	🔢 Number of Trades	💵 Avg Daily P&L
1.60	-27.87%	87	$+194.5K
↑ ≥1.0 investable	↑ Worst peak-to-trough loss	↑ 206 buy days / 364 total	↑ on active trade days

Dollar Impact

💸 Gross Profit	🛡️ Capital Preserved vs Buy&Hold	⛽ Est. Trading Fees	🏆 Best Model
$50.00M	$5.88M avoi…	$4350.0K	RandomForest
↑ Strategy gain in USD		↑ 0.1% per trade (Binance rate)	↑ selected for max return

📌 **IC Summary:** Starting from $50M**, the **RandomForest**strategygrewtheportfolioto**100.00M (+100.0%) over the 2024 test period. This compares to **$105.88M** (+111.8%) if we had simply held Bitcoin. The Sharpe Ratio of **1.60** demonstrates a risk-adjusted return suitable for the Investment Committee.

🔍 Data Pipeline Inspection

Raw Market Data | Engineered Features | Buy Signal Distribution

Google Trends (`gtrends`) — 0-100 scale. 100 = peak search interest. Fetched weekly via pytrends, interpolated to daily, **lagged by 1 day** to prevent look-ahead bias.

Date	open	high	low	close
2020-01-20 00:00:00	8704.6318	8745.5908	8560.4736	8657.6
2020-01-21 00:00:00	8658.9912	8755.7061	8544.5205	8745.8
2020-01-22 00:00:00	8744.2109	8792.9941	8636.7471	8680.
2020-01-23 00:00:00	8680.6504	8687.7471	8333.6377	8406.5
2020-01-24 00:00:00	8405.5674	8514.667	8266.8408	8445.4
2020-01-25 00:00:00	8440.1191	8458.4531	8296.2188	8367.8
2020-01-26 00:00:00	8364.4102	8602.4014	8325.498	8596.8
2020-01-27 00:00:00	8597.3086	8977.7266	8597.3086	8909.8
2020-01-28 00:00:00	8912.5244	9358.5898	8908.4473	9358.5
2020-01-29 00:00:00	9357.4707	9406.4316	9269.4678	9316.6

Date	open	high	low	close
2024-12-20 00:00:00	97484.6953	98098.9141	92175.1797	97755.9
2024-12-21 00:00:00	97756.1953	99507.1016	96426.5234	97224.7
2024-12-22 00:00:00	97218.3203	97360.2656	94202.1875	95104.9
2024-12-23 00:00:00	95099.3906	96416.2109	92403.1328	94686.2
2024-12-24 00:00:00	94684.3438	99404.0625	93448.0156	98676.0
2024-12-25 00:00:00	98675.9141	99478.75	97593.4688	99299.1
2024-12-26 00:00:00	99297.6953	99884.5703	95137.8828	95795.5
2024-12-27 00:00:00	95704.9766	97294.8438	93310.7422	94164.8
2024-12-28 00:00:00	94160.1875	95525.8984	94014.2891	95163.9
2024-12-29 00:00:00	95174.0547	95174.875	92881.7891	93530.2

🧮 Logistic Regression Formula

The complete log-odds equation. A **positive coefficient** means higher feature value → higher P(Buy).

	Feature	Coef
0	fear_greed	
1	rsi	
2	adx	
3	macro_pressure	
4	sp500_ret1	-0.035217
5	trend_score	0.024185

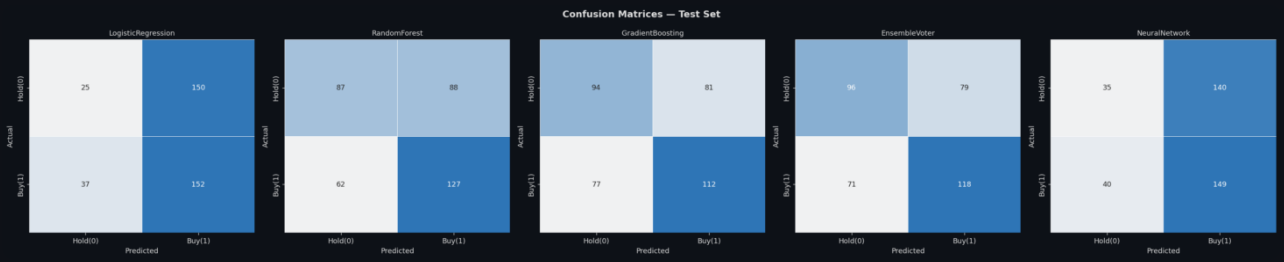
```
Log-Odds = 0.0004
   + 0.1774 * fear_greed
   − 0.1514 * rsi
   + 0.0897 * adx
   − 0.0455 * macro_pressure
   − 0.0352 * sp500_ret1
   + 0.0242 * trend_score
```

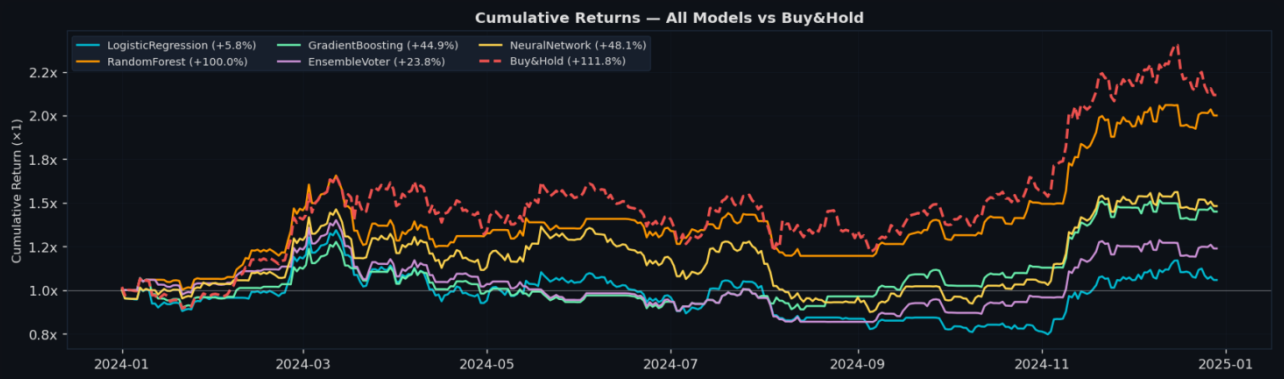
📊 Pipeline & Backtest Charts

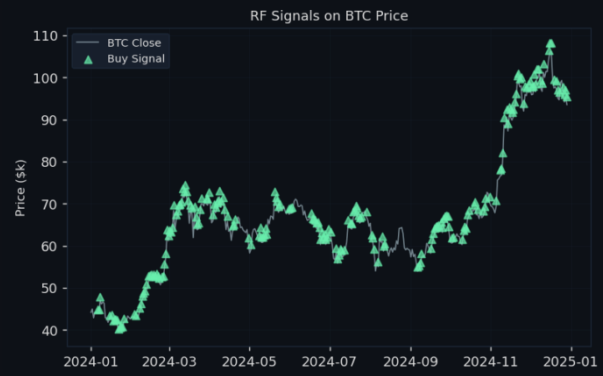
Strategy: Risk-Managed Long/Cash | Kill Switch: -5% daily loss | PROBA_CUTOFF: 49%

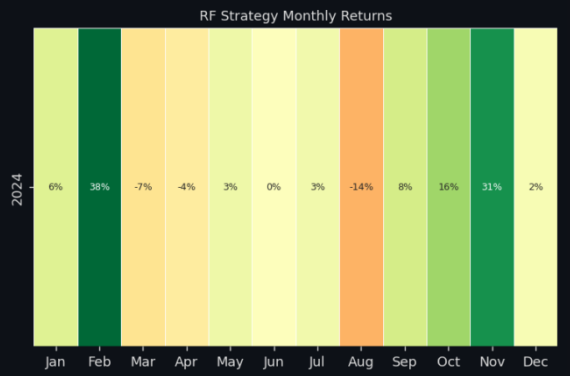
BTC Technical Indicator Overview — Full Period

Model Evaluation Metrics — Validation vs Test

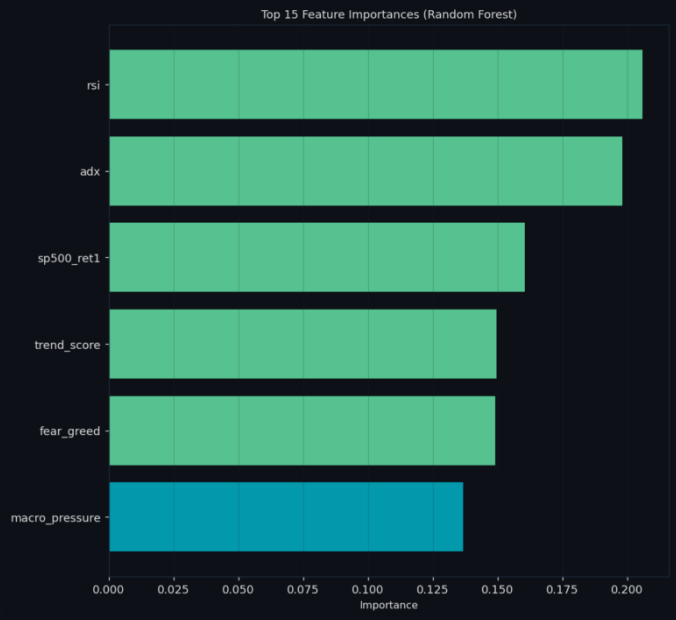


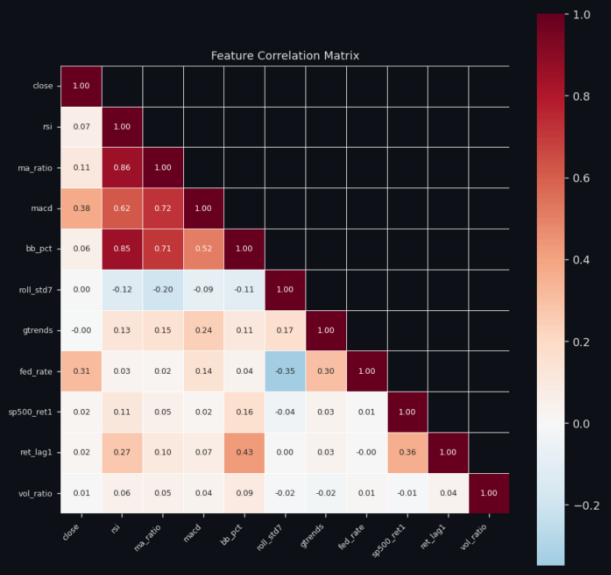






Feature Analysis





📈 Full Model Evaluation Metrics

	Model	Split	Precision	Recall	F1	Accur
0	LogisticRegression	Validation	0.505600	0.747300	0.603100	
1	LogisticRegression	Test	0.503300	0.804200	0.619100	
2	RandomForest	Validation	0.510200	0.686800	0.585500	
3	RandomForest	Test	0.590700	0.672000	0.628700	
4	GradientBoosting	Validation	0.522600	0.571400	0.545900	
5	GradientBoosting	Test	0.580300	0.592600	0.586400	
6	EnsembleVoter	Validation	0.512100	0.582400	0.545000	
7	EnsembleVoter	Test	0.599000	0.624300	0.611400	
8	NeuralNetwork	Validation	0.494800	0.791200	0.608900	
9	NeuralNetwork	Test	0.515600	0.788400	0.623400	